

# Intra-articular knee

## NOVA-MSF FLEX + NOVA-EXOS CORE • physician reference protocol

**PER KNEE****2 cc total**

FLEX 1 cc + EXOS CORE 1 cc

**BILATERAL****4 cc total**

2 cc per joint

**NEEDLE****22-25 G**

1.5 in, longer if indicated

**OBSERVATION****10-15 min**

post-treatment

A potential approach for intra-articular administration in patients presenting with symptomatic knee degeneration, osteoarthritis, chronic knee pain, meniscal degeneration, or cartilage wear.

### Indications

- Mild-to-moderate osteoarthritis
- Chronic knee pain
- Degenerative meniscal pathology
- Cartilage wear without complete joint collapse

### Contraindications

- Active infection or septic arthritis
- Uncontrolled coagulopathy
- Known product hypersensitivity
- Intra-articular corticosteroid within four weeks

### Procedure

**PRE-PROCEDURE**

1. Review patient history and available imaging.
2. Obtain informed consent.
3. Verify product preparation and integrity.
4. Position supine, knee flexed ~90°.
5. Prepare skin with chlorhexidine or povidone-iodine.

**POST-PROCEDURE — PATIENT INSTRUCTIONS**

- Avoid NSAIDs 48-72 h when clinically appropriate
- Avoid alcohol 48-72 h
- Limit strenuous lower-extremity activity 2-3 days
- Cold therapy as needed; resume rehab as tolerated

**INJECTION**

1. Aspirate effusion if present.
2. Confirm intra-articular placement.
3. Slowly administer 1 cc NOVA-MSF FLEX and 1 cc NOVA-EXOS CORE.
4. Apply sterile dressing after needle withdrawal.

Preferred: anterolateral or anteromedial approach. Alternative: superolateral with knee in extension.

**FOLLOW-UP & OUTCOME MEASURES**

Clinical assessment at **4-6 weeks**. Consider repeat treatment or adjunctive therapy at **8-12 weeks** based on provider assessment and response.

VAS PAIN

WOMAC

KOOS

FUNCTIONAL STATUS

ACTIVITY LEVEL

**REGULATORY**

CelluNOVA products are supplied for use by licensed healthcare professionals. CelluNOVA does not direct, prescribe, or control the practice of medicine, and nothing here guarantees a clinical outcome.

CHART DOCUMENTATION · EVALUATION VISIT

|                      |                                            |                            |               |
|----------------------|--------------------------------------------|----------------------------|---------------|
| PATIENT NAME         | DATE OF BIRTH                              | PATIENT ID / MRN           | DATE OF VISIT |
| EVALUATING PHYSICIAN | INITIAL · FOLLOW-UP<br>VISIT TYPE — CIRCLE | REFERRING PROVIDER, IF ANY | TIME IN / OUT |

**S SUBJECTIVE**

**Knee pain —**  
 CHIEF COMPLAINT

ONSET / DURATION

SEVERITY — VAS 0-10

QUALITY / RADIATION

AGGRAVATING FACTORS

RELIEVING FACTORS

**PT · NSAIDs · corticosteroid inj · hyaluronic acid · bracing · arthroscopy**  
 PRIOR TREATMENT — CIRCLE ALL TRIED · ADD DATES

stairs · kneeling · walking distance · sleep · work · sport  
 FUNCTIONAL IMPACT

fever · chills · weight loss · neuro changes — note positives  
 PERTINENT ROS

MEDICATIONS — INCL. ANTICOAGULANTS, IMMUNOSUPPRESSANTS

ALLERGIES · RELEVANT PMH — CANCER, INFECTION, PREGNANCY

**O OBJECTIVE**

BP

HR

TEMP

WEIGHT

GENERAL APPEARANCE

Inspection — effusion, alignment, quadriceps bulk

Palpation — joint line, patellofemoral tenderness

ROM — flexion / extension, degrees

Stability — Lachman, varus/valgus, McMurray

Gait and squat tolerance

IMAGING / STUDIES REVIEWED — MODALITY, DATE, FINDINGS

BASELINE — VAS /10

BASELINE — WOMAC

BASELINE — KOOS

**A ASSESSMENT**

**Symptomatic knee OA · chronic knee pain · degenerative meniscus**  
 WORKING DIAGNOSIS — CONFIRM OR AMEND

ADDITIONAL DIAGNOSES / CODES

☐ Appropriate candidate — indications met, contraindications absent

☐ Alternatives discussed, including conservative care and doing nothing

☐ Non-FDA-approved status explained; no outcome promised

☐ Patient questions answered; understanding confirmed

**P PLAN**

**Proposed treatment.** Intra-articular injection, affected knee(s): NOVA-MSF FLEX 1 cc + NOVA-EXOS CORE 1 cc per knee, 22-25 G, landmark or ultrasound guidance. Observation 10-15 min. NSAID and alcohol hold explained.

☐ Informed consent (CN-PC) reviewed and signed

☐ Procedure performed today — see CN-OP-KNEE · or scheduled (date right)

☐ Fees, non-coverage by insurance, and refund policy reviewed

☐ Follow-up: assessment at 4-6 wk; repeat consideration 8-12 wk

PHYSICIAN SIGNATURE

PRINTED NAME & LICENSE NUMBER

PROCEDURE DATE, IF SCHEDULED

PATIENT NAME

DATE OF BIRTH

PATIENT ID / MRN

DATE OF PROCEDURE

PERFORMING PHYSICIAN

Symptomatic knee OA · chronic knee pain · degener...

PRE-PROCEDURE DIAGNOSIS — CONFIRM OR AMEND

START TIME

ASSISTING STAFF

PRE-PROCEDURE CHECKLIST

Check each box before the first draw

- ☐ Informed consent (CN-PC) signed and on file; SOAP note (CN-SN-KNEE) complete
- ☐ Screening reviewed — no contraindications present
- ☐ Time-out performed — correct patient, site, product, route
- ☐ Five checks read aloud with second staff: ID · product & lot · expiry · volume & route · consent
- ☐ Baseline outcome measures recorded: VAS pain · WOMAC · KOOS
- ☐ Site marked and prepped — chlorhexidine / povidone-iodine (circle); chain-of-custody log initialed

BP · HR · TEMP

agent · concentration · volume — write NONE i...

BP · HR · TEMP

PRE-PROCEDURE VITALS

LOCAL ANESTHETIC

POST-OBSERVATION VITALS

PRODUCTS ADMINISTERED

Per knee; double volumes for bilateral treatment

| PRODUCT & TARGET                                 | PLANNED     | LOT | EXPIRY | GIVEN |
|--------------------------------------------------|-------------|-----|--------|-------|
| NOVA-MSC FLEX<br>Intra-articular, affected knee  | 1 cc / knee |     |        |       |
| NOVA-EXOS CORE<br>Intra-articular, affected knee | 1 cc / knee |     |        |       |

TECHNIQUE

Position: supine, knee flexed ~90° (extension for superolateral)

- Approach: anterolateral or anteromedial; alternative superolateral in extension.
- Aspirate effusion if present; confirm intra-articular placement before injecting.
- Administer FLEX then EXOS CORE slowly; sterile dressing after withdrawal.

LEFT · RIGHT · BILATERAL — circle

SITE / SIDE

LANDMARK · ULTRASOUND — circle

GUIDANCE

22-25 G, 1.5 in

NEEDLE USED

target 10-15 min

OBSERVATION — MINUTES

NARRATIVE — FINDINGS, ASPIRATE, DEVIATIONS FROM PROTOCOL, PATIENT RESPONSE

POST-PROCEDURE & DISPOSITION

Tolerated: WELL · WITH EVENTS — circle

Discharged: STABLE · AMBULATORY · WITH ESCORT — circle

☐ Patient note CN-PN-KNEE reviewed and given at discharge

COMPLICATIONS / ADVERSE EVENTS — WRITE "NONE" IF NONE

Any adverse event also requires the adverse-event response steps in this book.

Follow-up plan. Clinical assessment at 4-6 weeks; consider repeat treatment or adjunctive therapy at 8-12 weeks based on response.

FOLLOW-UP SCHEDULED FOR

PHYSICIAN SIGNATURE

PRINTED NAME & LICENSE NUMBER

DATE & END TIME

Today you received an injection of cell and exosome products into your knee joint. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

**WHAT TO EXPECT**

Changes are gradual — do not judge the result in the first days

**FIRST 72 HOURS**

Soreness, warmth, or mild swelling in the knee is common and usually settles in two to three days. A short flare of your usual pain can happen.

**WEEKS 1-4**

Ease back into normal activity as comfort allows and restart physical therapy or home exercises as advised.

**WEEKS 4-12**

This is when progress is assessed. Keep your follow-up visit so we can measure how the knee is responding.

**DO**

- Take it easy the rest of today; gentle walking is fine
- Use cold packs 10-15 minutes at a time for comfort
- Take acetaminophen (Tylenol) for soreness if needed
- Resume physical therapy or home exercises as advised

**AVOID**

- Anti-inflammatory pain relievers (ibuprofen, naproxen, aspirin for pain) for 72 hours
- Alcohol for 72 hours
- Squatting, impact, or strenuous lower-body exercise for 2-3 days
- Hot tubs or soaking the injection site for 24 hours

**CALL US RIGHT AWAY IF**

For an emergency, call 911 first

- The knee becomes increasingly hot, red, and swollen rather than better
- Fever above 101 °F or shaking chills
- Calf pain or swelling, or shortness of breath
- Severe pain not controlled by acetaminophen and cold packs

CLINIC PHONE

AFTER-HOURS PHONE

**NOTES FROM YOUR VISIT**